2026Year 2nd Semester Syllabus

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Course Title	SOFTWARE ENGINEERING	Course Code-Section	CAS3106-01
Credit/Time/ Experiment,Lab,Practical Technique Time	3/Tue7,Thu8,9	Department	School of Computing
Time	Tue7,Thu8,9	Location	EngHD504
Exam Date & Time	Midterm exam	Final exam	
Class Language	English	Evaluation Type	Absolute evaluation

Instructor's Profile	Name	Kim Gyuyeong	Contact Information	Telephone	
	Department	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING		Mail	gyuyeong.kim@yonsei.ac.kr
	Office			Interview information	

TA's Name & Contact Information	Name		Contact Information	Telephone	

Course Description Brief Introduction of the Course	소프트웨어 공학은 컴퓨터 소프트웨어의 개발에 필요한 이론, 방법 그리고 도구를 다루는 학문이다. 소프트웨어 개발 단계는 요구사항 정의, 시스템과 소프트웨어 설계, 구현과 단위 (unit) 테스트, 통합 및 시스템 테스트, 운영 및 관리로 이루어진다. 본 과목에서는 각각의 개발 단계에 사용되는 여러 가지 방법에 대해서 세부적으로 다룬다.			
Course Goals	1.	Korean	소프트웨어공학의 개념과 개발 생명주기(SDLC)를 설명할 수 있다.	20%
		English	Explain the concepts of software engineering and the SDLC.	
	2.	Korean	버전관리(Git/GitHub)·컨테이너(Docker) 등 개발·협업 도구를 활용할 수 있다.	20%
		English	Use development and collaboration tools such as Git/GitHub and Docker.	
	3.	Korean	대규모 시스템의 아키텍처를 설계하고 성능·확장성·일관성 트레이드오프를 분석할 수 있다.	20%
		English	Design large-scale system architectures and analyze performance, scalability, and consistency trade-offs.	
	4.	Korean	객체지향 설계 원칙(OOP·SOLID)과 핵심 디자인 패턴을 적용할 수 있다.	20%
		English	Apply object-oriented design principles (OOP, SOLID) and key design patterns.	
	5.	Korean	클린코드·코드리뷰로 구현 품질을 높이고, 테스트를 설계·수행한다.	20%
		English	Improve code quality through clean code and code review, and design, run,and validate tests	

Core Competencies	The total measurable competencies must be 100%. Each course objective should set the competency as 25%. The core and major competencies should equal at least 50%.		
	협동 50%	프로그래밍능력 40%	문제해결능력 10%
Sub-Competencies/Learning Unit1			
Sub-Competencies/Learning Unit2			
Sub-Competencies/Learning Unit3			
Core Competencies(Liberal Arts)Major competency(	**Must reflect the interrelationship between core competencies (elective courses) and major competencies (major studies).**		
협동			
Sustainable Development Goals			

Average Recommended Amount of Learning per	Average Reading Volume		Average amount of writing(Based on A4)	

Course Methods (%) Total Amount 100	Lecture	Practice Training	Presentation	Dabate	Team Project
	100%	0%	0%	0%	0%
Course Methods 2 Select Relevant Items	PBL Subject	Capstone Design	CBL, Social Innovation Course	Flipped Classroom	Work Experience,Internsh

Grading Policy(%) Total Amount 100 Free Input for Other Information	Midterm exam	Final exam	Quiz	Individual Assignment	Team Assignment	Attendance	Others
	45%	50%	0%	0%	0%	5%	0%

Assignment/ Report, Project Guide	Title of Assignment/Project Name, and Method of Filling Out	Submission Deadline	Type of Submission and Method
Prerequisite	Data structure, Object-Oriented Programming	Online Course Address	

Course Material	Course Material Name	Author	Publisher	Publish Year	ISBN
주교재	Software engineering	Sommerville, Ian	Pearson Education, Inc	2016 / 10th ed	0133943038
참고자료	no single book covers everything ; useful for the overall process				

Main Learner Precautions	
Attatchment	

Weekly Plan

week	Period	Weekly Topic & Contents	Remarks
1	2026-09-01 2026-09-07	Introduction to Course, Introduction to Software Engineering	(9.1.) Fall semester classes begin (9.3. - 9.7.) Course add and drop period
2	2026-09-08 2026-09-14	Requirements & Design, Implementation, Testing, Deployment & Maintenance	
3	2026-09-15 2026-09-21	Process Models, DevOps, Cloud, and Open-Source Licenses	
4	2026-09-22 2026-09-28	Version Control: Git / GitHub, Containers & Docker + Architecture Design Overview	(9.24. - 9.25.) Chuseok Holidays 09.24 추석, 09.25 추석, 09.26 추석
5	2026-09-29 2026-10-05	Load Balancer, Cache, and CDN, API Gateway, Storage, Message Queue, Performance & Scalability	(10.5.) Alternative Holiday of the National foundation Day 10.03 개천절, 10.05 대체공휴일(개천절)
6	2026-10-06 2026-10-12	Consistency, Availability, CAP, and Scaling, UML Diagrams	(10.7.) First third of the semester ends (10.9.) Hangul Proclamation Day 10.09 한글날
7	2026-10-13 2026-10-19	Architecture Styles, Four Pillars of OOP	(10.13. - 10.15.) Course withdrawal period
8	2026-10-20 2026-10-26	Midterm Exam	(10.20. - 10.26.) Midterm Examinations
9	2026-10-27 2026-11-02	SOLID: SRP, OCP, and LSP, SOLID: ISP and DIP + DRY, KISS, and YAGNI	(10.27. - 10.29.) Application Period for S/U evaluation
10	2026-11-03 2026-11-09	Design Patterns Overview, Creational Patterns	
11	2026-11-10 2026-11-16	Structural Pattern: Adapter, Behavioral Pattern: Observer	(11.13.) Second third of the semester ends
12	2026-11-17 2026-11-23	Behavioral Pattern: Strategy, Clean Code, Style Guide, and Code Review	
13	2026-11-24 2026-11-30	Testing Fundamentals: Error, Fault, Failure, Types, and Principles, Test Levels, Oracle, and Coverage Concepts	
14	2026-12-01 2026-12-07	Test Design, Coverage, and Static Analysis	
15	2026-12-08 2026-12-14	Self-Study Week	(12.8. - 12.14.) Self-study
16	2026-12-15 2026-12-21	Final Exam	(12.15. - 12.21.) Final Examinations

Attendance Requirement

• Students who are absent for more than one-third of the total class hours will receive an F or NP grade regardless of examination results.
• Courses without midterm and/or final examinations will hold regular classes during the examination periods.

Support for Students with Disabilities(SWDs)

• SWDs can request accommodations related to lectures, assignments, or tests by contacting the course professor at the beginning of semester. However, accommodations may vary depending on the essentiality of lecture and discretion of professors.

[Lecture]
- Visual Impairment: alternative, braille, enlarged reading materials, note-taker
- Physical Impairment: alternative reading materials, access to classroom, note-taker, assigned seat
- Hearing Impairment: note-taker/stenographer, recording lecture
- Intellectual Disability/Autism: note-taker

[Assignments and Test]
- Visual/Physical/Hearing Impairment: (reasonable) extra days for submission, alternative type of assignment, extended test time, alternative type of test, arranging separate test room, and proctors, test ghostwriter
- Intellectual Disability/Autism: (reasonable) extra days for submission, alternative type of assignment

Safety Precautions

• Students required to complete laboratory safety training must finish the online training before the semester begins and submit the certificate at the first class session. Failure to do so may result in restricted class participation.

• Students in physical activity courses should complete appropriate warm-up exercises before class. Those with cardiovascular or chronic respiratory conditions are advised to consult a physician before participating.

Course Administration

• Notices regarding class cancellations, make-up classes, classroom changes, and other course-related matters may be posted on LearnUs. For updates, please check LearnUs regularly.